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Predicting Protein Crystallization Conditions

Data Acquisition, Preprocessing, Exploratory Data Analysis

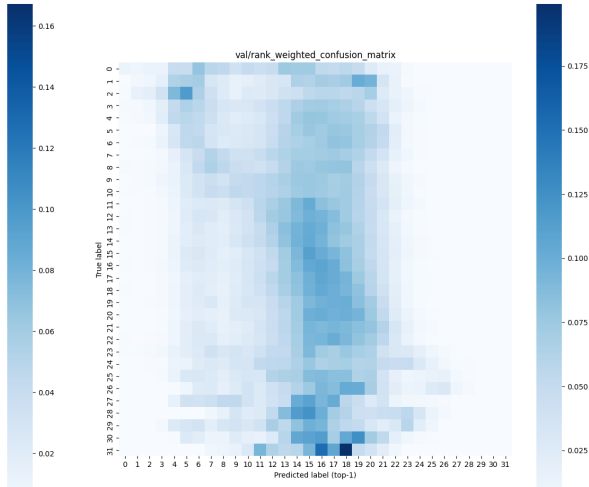
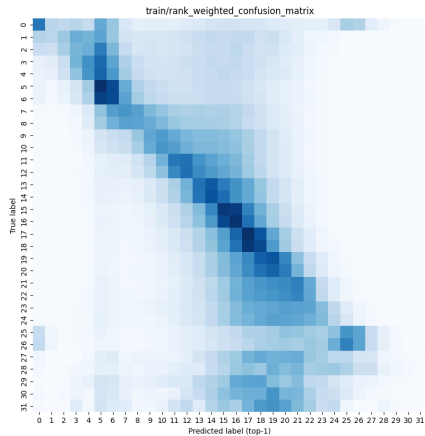
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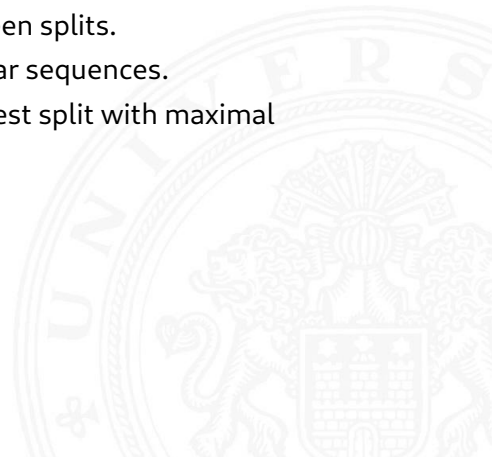
Maschinelles Lernen in der Bioinformatik

February 10, 2026



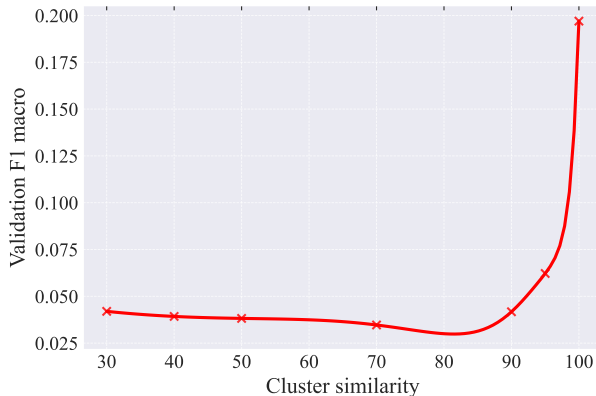
- ▶ More thorough comparison of model
- ▶ test joinedTail, separateTails, RF and XGBoost
- ▶ XGBoost wins by far so focusing on that
- ▶ Finetune large language model to improve parsing and handling of more obscure free text descriptions → not much better than parsing in the end (**on hold**)
- ▶ Implement Condition Encoding using SMILES and Morgan fingerprints → worked to a degree using HDBSCAN majority of conditions can be clustered meaningfully, however 25% were outliers (**in progress**)
- ▶ implement sequence similarity based clustering

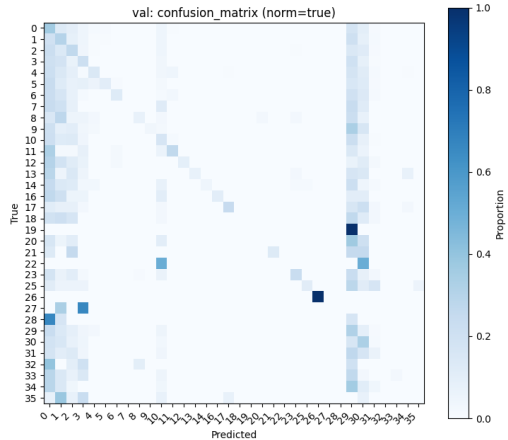
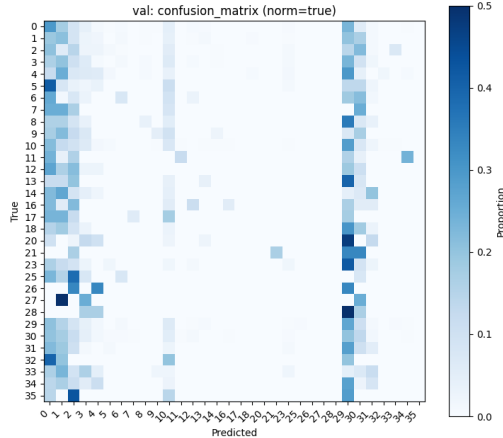
- ▶ Sequences are clustered by similarity to group homologous proteins.
- ▶ Train/validation splits are performed at the cluster level.
- ▶ Clustering thresholds control homology between splits.
- ▶ Higher thresholds yield easier splits with similar sequences.
- ▶ Coarser clustering (e.g. 30%) creates the hardest split with maximal dissimilarity.



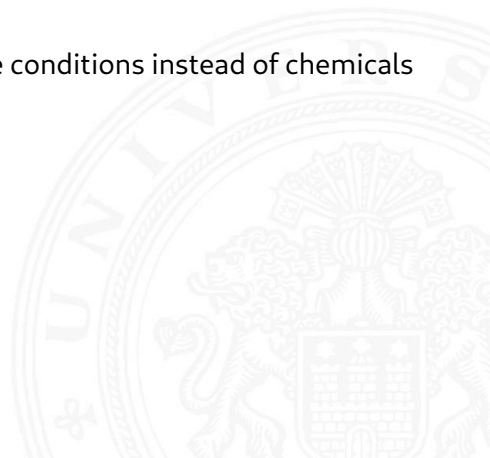
- ▶ Different levels tested
30%, 40%,
50%, 70%, 90%, 95%, 100%
- ▶ Model loses
generalization power
extremely quickly
- ▶ likely performs nearest
neighbor in high
dimensional
embedding space

Performance of model for different cluster similarity thresholds





- ▶ A lot of papers do not report sequence similarity based clustering for building a validation split. What would be a valid split?
- ▶ Optimize PU learning for XGBoost
- ▶ Improve condition clustering and predict entire conditions instead of chemicals



Bekker, Jessa and Jesse Davis (Apr. 2020). "Learning from positive and unlabeled data: a survey". In: *Machine Learning* 109.4. arXiv:1811.04820 [cs], pp. 719–760. ISSN: 0885-6125, 1573-0565. DOI: 10.1007/s10994-020-05877-5. URL: <http://arxiv.org/abs/1811.04820> (visited on 01/27/2026).